

9.5.20.

- a. How many distinguishable ways can the letters of the word *MILLIMICRON* be arranged in order?
- b. How many distinguishable orderings of the letters of *MILLIMICRON* begin with *M* and end with *N*?
- c. How many distinguishable orderings of the letters of *MILLIMICRON* contain the letters *CR* next to each other in order and also the letters *ON* next to each other in order?
- a. How many distinguishable ways can the letters of the word *MILLIMICRON* be arranged in order?

There're eleven letters of the word *MILLIMICRON*. Since the words have be arranged in a distinguishable way, and there're nondistinct letters, we must choose lock down the positions for those letters, before arranging the distinct ones. There're two *M*'s, three *I*'s, and two *L*'s. Hence:

$$\begin{aligned} M &: \binom{11}{2} \text{ choose 2 of 11 positions} \\ I &: \binom{9}{3} \text{ choose 3 of the 9 remaining positions} \\ L &: \binom{6}{2} \text{ choose 2 of the 6 remaining positions} \end{aligned}$$

Since *C*, *R*, *O*, *N* are distinct, the remaining four letters can be arranged in $4!$ ways.

Solution: a. There're $\binom{11}{2} \times \binom{9}{3} \times \binom{6}{2} \times 4!$ distinguishable arrangements.

b. How many distinguishable orderings of the letters of *MILLIMICRON* begin with *M* and end with *N*?

Since the ordering must begin with *M* and end with *N*, these two positions are fixed. This leaves 9 positions containing one *M*, three *I*'s, two *L*'s, and the distinct letters *C*, *R*, and *O*.

$$\begin{aligned}
M &: \binom{9}{1} \text{ choose 1 of 9 positions} \\
I &: \binom{8}{3} \text{ choose 3 of the 8 remaining positions} \\
L &: \binom{5}{2} \text{ choose 2 of the 5 remaining positions}
\end{aligned}$$

Since C , R , O are distinct, the remaining three letters can be arranged in $3!$ ways.

Solution: b. There're $\binom{9}{1} \times \binom{8}{3} \times \binom{5}{2} \times 3!$ distinguishable arrangements, if the orderings must begin with M and end with N .

c. How many distinguishable orderings of the letters of *MILLIMICRON* contain the letters CR next to each other in order and also the letters ON next to each other in order?

Since CR and ON are required pairs, and their internal order is already fixed, we now effectively have 9 objects. There're again two M 's, three I 's, and two L 's.

$$\begin{aligned}
M &: \binom{9}{2} \text{ choose 2 of 9 positions} \\
I &: \binom{7}{3} \text{ choose 3 of the 7 remaining positions} \\
L &: \binom{4}{2} \text{ choose 2 of the 4 remaining positions}
\end{aligned}$$

Since CR and ON are distinct blocks, the remaining two blocks can be arranged in $2!$ ways.

Solution: c. There're $\binom{9}{2} \times \binom{7}{3} \times \binom{4}{2} \times 2!$ distinguishable arrangements.